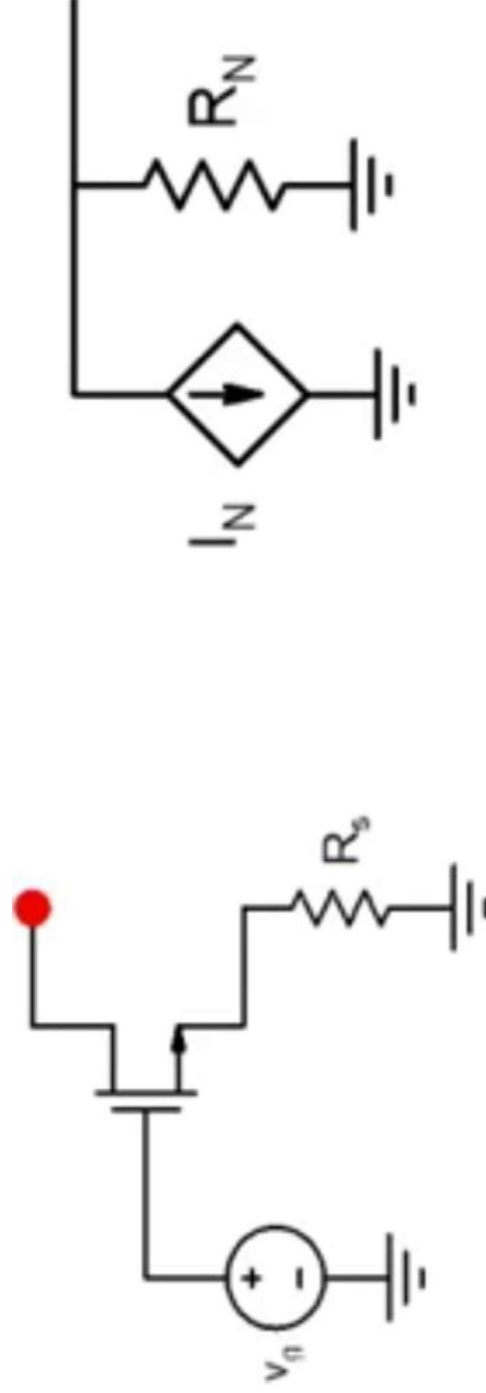


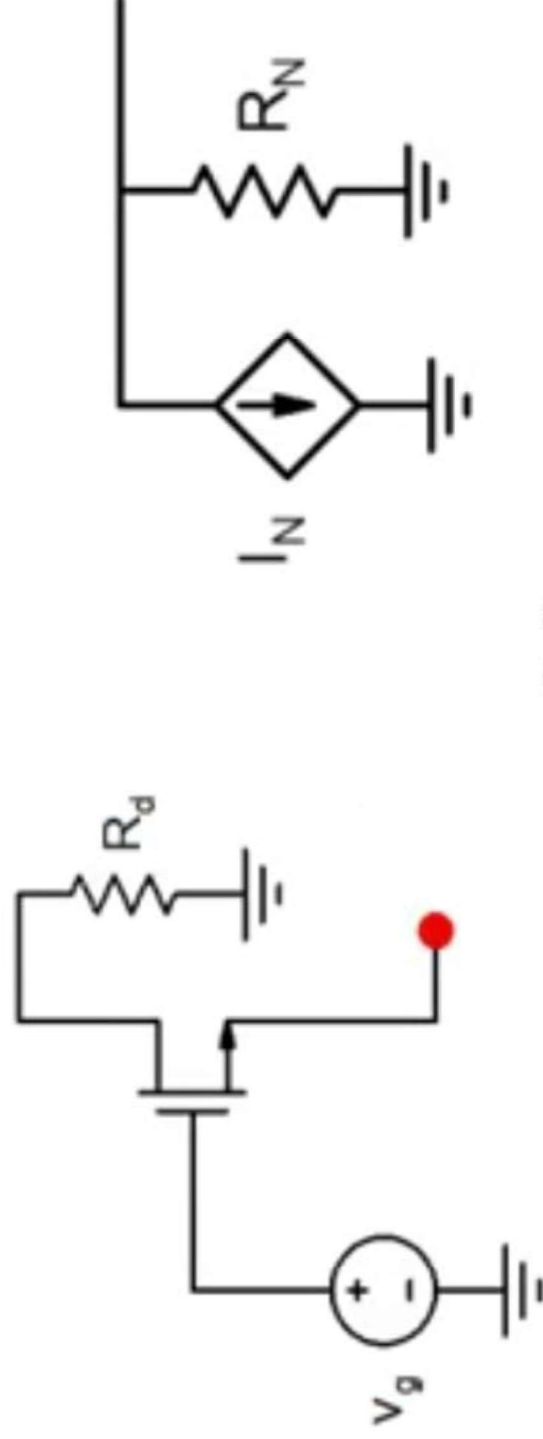
## Useful Results



$$i_n \cong \frac{g_m v_g}{1 + (g_m + g_{mb})R_S}$$

$$R_n \cong r_o \times \{1 + (g_m + g_{mb})R_S\}$$

## Useful Results



$$i_n = -\frac{g_m v_g}{1 + R_d / r_o}$$

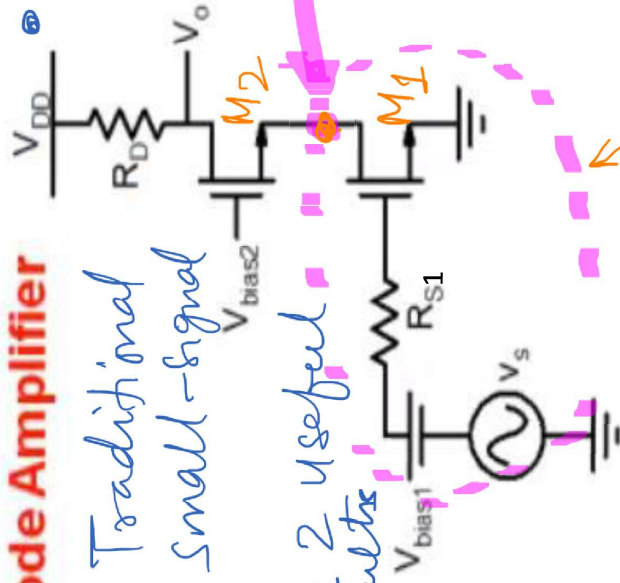
$$R_{in} \cong \frac{1 + R_d / r_o}{g_m + g_{mb}}$$

## Cascode Amplifier

Let's calculate the Gain of this Amplifier

Method-1: Traditional way  $\rightarrow$  Small-signal

Method-2: Use 2 useful Results

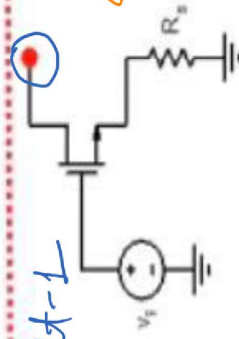


new eq

$$i_n \approx g_{m1} V_s$$

$$R_{th} \approx R_{S1}$$

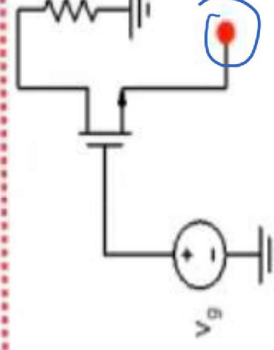
useful Result-1



$$i_n \approx \frac{g_m v_g}{1 + (g_m + g_{mb}) R_S}$$

$$R_{th} \approx r_o \times \{1 + (g_m + g_{mb}) R_S\}$$

useful Result-2



$$i_n = -\frac{g_m v_g}{1 + R_d / r_o}$$

$$R_{th} \approx \frac{1 + R_d / r_o}{g_m + g_{mb}}$$

# Cascode Amplifier

Built Norton eq at this node  $\rightarrow$

The whole ckt looking from this node is a current source  $\rightarrow$  parallel with  $r_{o1}$

Now, Built Norton eqn at  $(V_o)$   
 $\rightarrow$  Find the Norton current  $\rightarrow$   $(I_N)$

$\rightarrow$  Source of  $I_N$  is  $g_{m1}V_s$

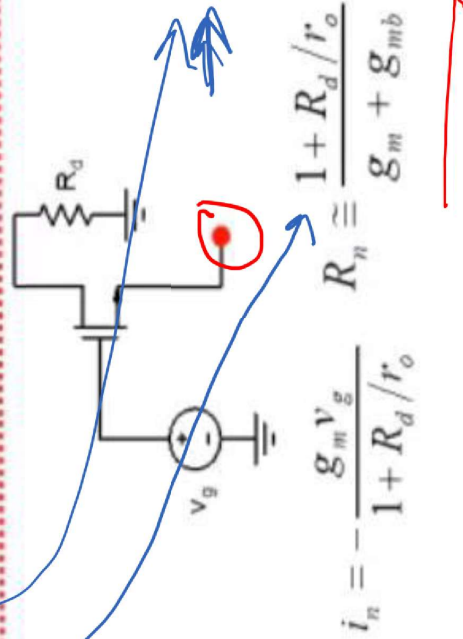
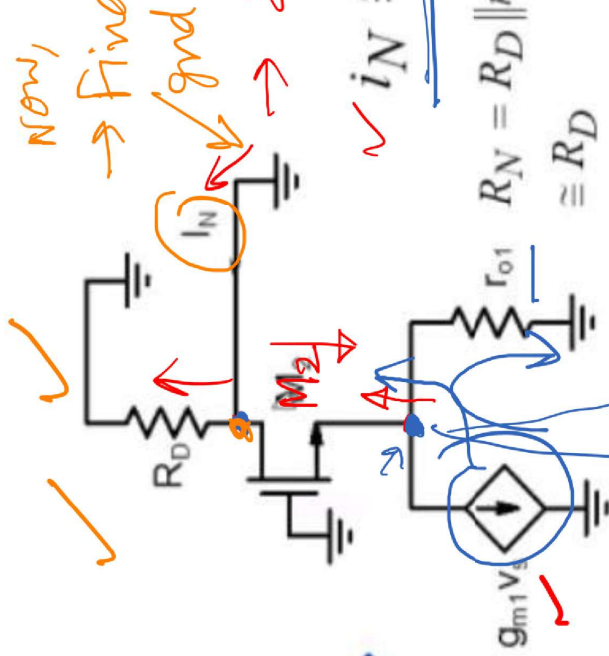
$\rightarrow$  This current  $(g_{m1}V_s)$  has

two choices  $\rightarrow$   $r_{o1}$   $\rightarrow$   $M_2$

$\rightarrow$  flow into  $M_2$  and come out as  $I_N$

what is Resistance when we look into  $M_2$

$$R_n = \frac{1}{g_{m2} + g_{mb2}}$$



$$i_n = -\frac{g_m V_g}{1 + R_d / r_o} \quad R_n \approx \frac{1 + R_d / r_o}{g_m + g_{mb}}$$

$$i_n \approx \frac{g_m V_g}{1 + (g_m + g_{mb}) R_S}$$

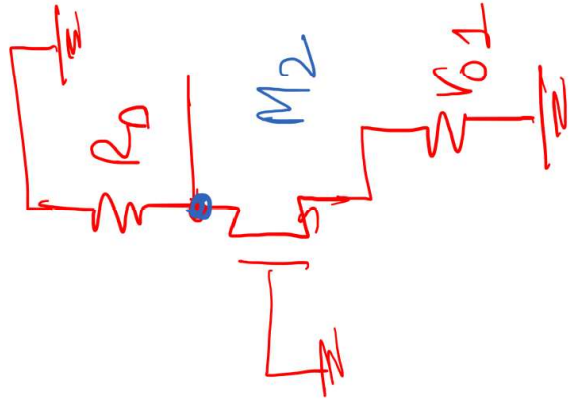
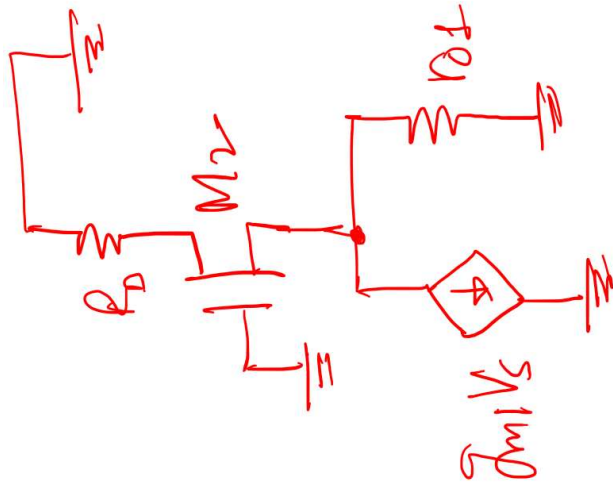
$$R_n \approx r_o \times \{1 + (g_m + g_{mb}) R_S\}$$

→ Remember: The order of  $V_D$  and resistance looking into source  
↓  $n M_2$  ↓  $n$

→ Therefore, the current will flow into  $M_2$

$$I_N \approx g_{m1} V_S$$

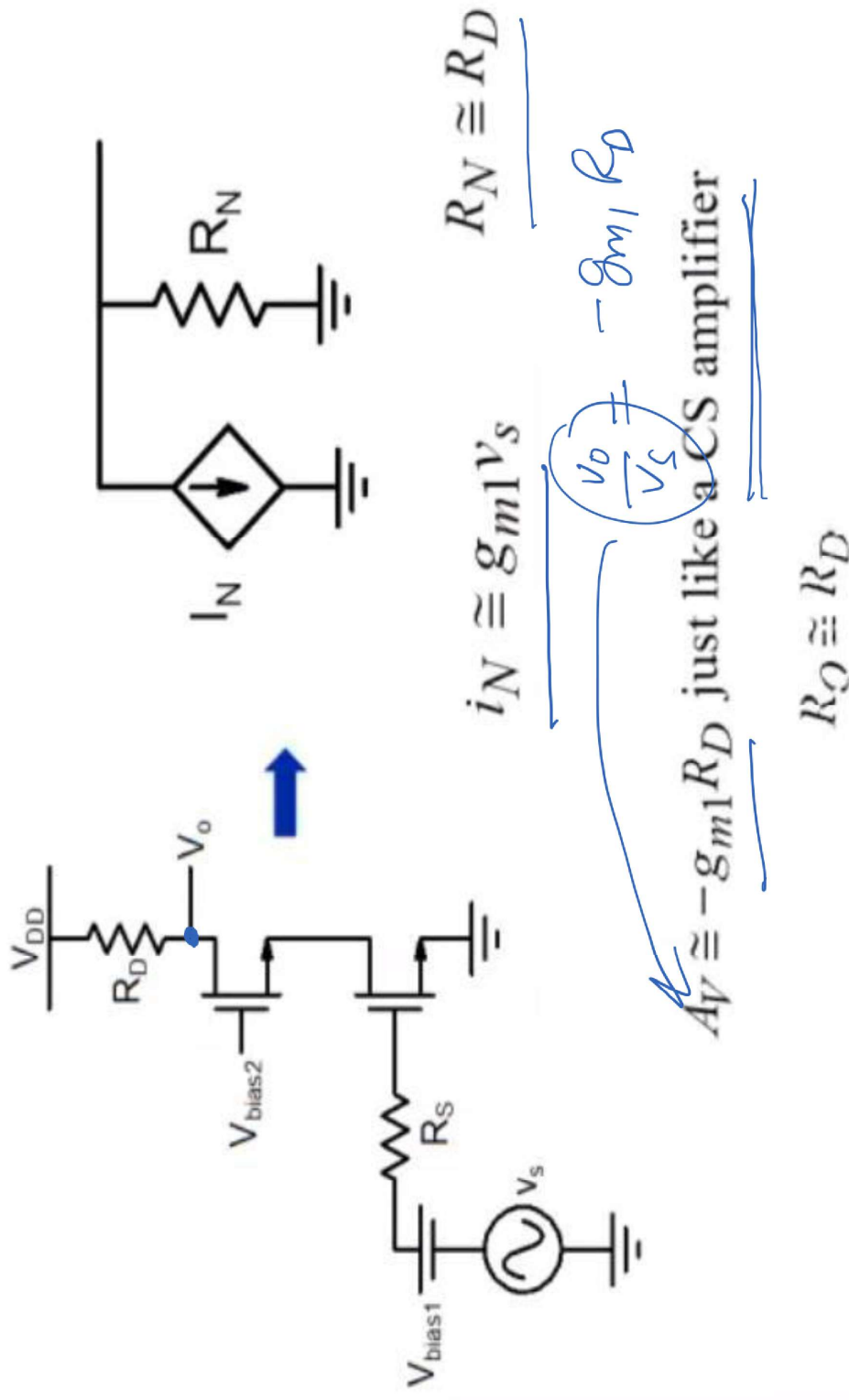
→ Now, let's find the Norton resistance at that node  
• forget the ground  
→ Resistance looking up →  $R_D$   
→ Resistance looking down → short all sources



$$R_N \approx R_D \parallel r_{o2} (1 + (g_{m2} + g_{mb2}) r_{o1})$$

$$R_N \approx R_D$$

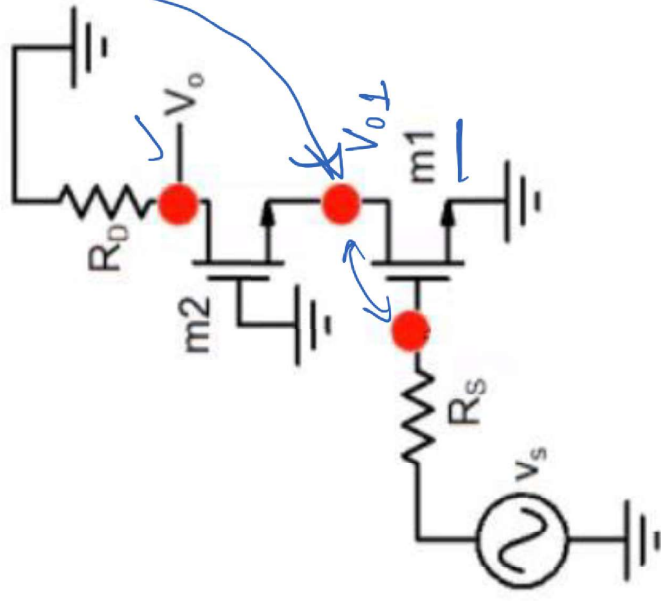
## Cascode Amplifier



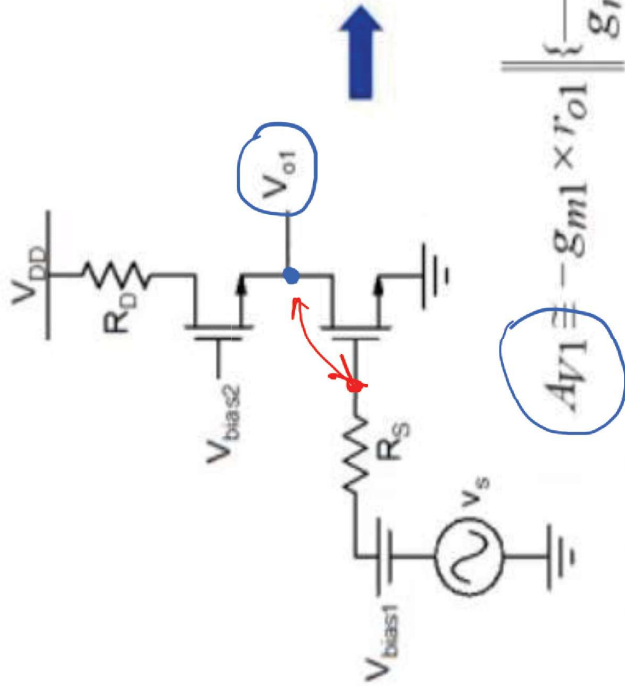


## Cascode Amplifier: Frequency Response

- $G_{\text{ain}}(M_1)$  will suffer from Miller effect
- Let's build a gain at  $V_{o1}$
- Build Norton eqn.



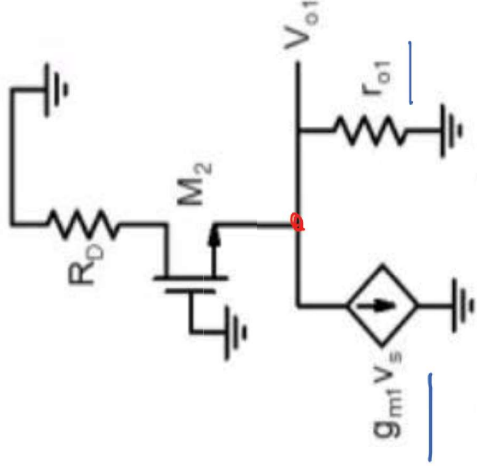




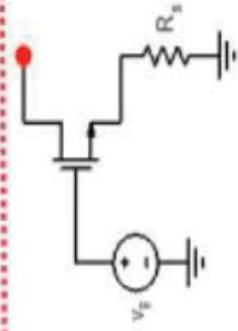
$$A_{V1} \cong -g_{m1} \times r_{o1}$$

$$i_n \cong \frac{g_m v_g}{1 + (g_m + g_{mb}) R_S}$$

$$R_{in} \cong r_o \times \{1 + (g_m + g_{mb}) R_S\}$$



$$\left\{ \frac{1}{g_{m2} + g_{mb2}} \left( 1 + \frac{R_D}{r_{o2}} \right) \right\} \cong - \frac{g_{m1}}{g_{m2} + g_{mb2}}$$



$$i_n = - \frac{g_m v_g}{1 + R_d / r_o}$$

$$R_n \cong \frac{1 + R_d / r_o}{g_m + g_{mb}}$$

let's solve this